Egg consumption and the risk of diabetes in adults, Jiangsu, China.

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BACKGROUND: Although egg consumption has been associated with elevated plasma levels of cholesterol and triglyceride and with risk of cardiovascular disease in some populations, epidemiologic studies on egg consumption and the risk of diabetes are extremely sparse, particularly in the Chinese population. METHOD: Data from a household survey in the year 2002 among 2849 adults aged ≥20 y from a nationally representative sample in Jiangsu Province, China, were used. Dietary information was assessed by a validated food frequency questionnaire and 3 d weighed food records. Fasting blood specimens were collected. RESULTS: After the adjustment for age, total calorie intake, education, smoking, family history of diabetes, and sedentary activity, egg consumption was significantly and positively associated with diabetes risk, particularly in women. The odds ratios (OR) (95% CI) of diabetes associated with egg consumption <2/wk, 2-6/wk, and ≥1/d in the total sample were 1.00, 1.75, 2.28 (1.14-4.54), respectively (P for trend 0.029). Corresponding ORs (95% CI) in women were 1.00, 1.66, and 3.01 (1.12, 8.12), respectively (P for trend 0.022). Additional adjustment of body mass index attenuated the association, but it remained significant. There was a similar, however, not statistically significant association in men. In addition, plasma triglyceride and total cholesterol levels were significantly higher in women who consumed ≥2 eggs/wk than those who consumed eggs less often. CONCLUSION: Egg consumption was positively associated with the risk of diabetes among the Chinese, particularly in women.